



Department of Electronic and Electrical Engineering
UNIVERSITY COLLEGE LONDON
TORRINGTON PLACE LONDON WC1E7JE

COMMERCIAL IN CONFIDENCE

Ballistics Test Report

Lead No. 3 shot, 36g load (0.030" choke)

by David Compton

Statistical Aerodynamics Group

**This report is issued strictly on the condition
that it is not to be used for publicity purposes**

Table of Contents

1. Test Specification

1.1 Ammunition

1.1.1 Supplied by Manufacturer

1.1.2 Range

1. Test Specification

1.1	Ammunition	2
1.2	Gun	2
1.3	Atmospheric Conditions	2
1.4	Ranges tested	3

1.1.2 Measured at UCL (mean values)

2. Results

2.1	Results Summary	4
2.2	Shot Cloud Profiles	7
2.3	Patterns	8

1.2 Gun

Gun Gauge:

Gun:

Barrel length:

Bore:

Chamber length:

Cartridge type:

Cartridge:

1.3 Atmospheric Conditions

1.3.1 Field data

Temperature:

Pressure:

Relative humidity:

Lead No. 3 shot, 36g load (0.030" choke)

1. Test Specification

1.1 Ammunition

1.1.1 Supplied by Manufacturer

Bore :	12
Length :	70 mm 2 ³ / ₄ "
Load :	36 g 1 ¹ / ₄ oz
Shot :	3
Wad :	Plastic
Material :	Lead
Batch no :	-

1.1.2 Measured at UCL (mean values)

Load per cartridge :	35.8 g
Pellets per cartridge :	192
Powder mass per cartridge :	1.53 g
Pellet mass :	187 mg
Pellet diameter :	3.15 mm
Velocity (at 2.5m from muzzle) :	363 m/s

1.2 Gun

Gun Gauge :	12 bore
Gun :	Proof Barrel (LPH)
Barrel length :	29 "
Bore :	18.4 mm
Chamber length :	70 mm, 2.75 "
Choke type :	Multi-choke (Teague)
Choke :	0.030 "

1.3 Atmospheric Conditions

1.3.1 Field trials

Temperature :	23 °C
Pressure :	1006 mb
Relative humidity :	56.5 %

APPENDIX B

Wind speed : 1 m/s
Wind direction (from direction of fire) : 225°

1.3.2 Storage Conditions prior to testing

Temperature : 28 °C
Relative humidity : 49 %

1.4 Ranges tested

Range : 20 - 50 m
Number of Ranges : 7
Increments : 5 m
Number of rounds per range : 10

Range (m)	Flight Time (ms)		Velocity (m/s)		Energy per Pellet (J)		Shot Cloud Length (m)
	Leading Edge	Trailing Edge	Leading Edge	Trailing Edge	Leading Edge	Trailing Edge	
20	80.6	89.4	233	220	5.0	5.0	1.4
25	78.7	89.3	245	220	5.6	4.9	2.3
30	76.4	111.3	242	213	5.5	4.5	2.9
35	126.0	176.2	223	198	4.6	3.7	3.7
40	147.3	182.4	206	185	4.0	3.3	3.6
45	164.4	180.3	197	173	3.4	3.0	3.9
50	171.3	229.0	180	163	3.0	2.5	4.1

Figure B1: Shows the flight times for the main section of the cloud, and the corresponding outputs generated by the analysis program for Lead No. 3 shot, 3kg lead (0.002" dia).

Figures B2, B3, B4, B5 show the variation of flight time, velocity, energy per pellet, and shot cloud length with range, for Lead No. 3 shot, 3kg lead (0.002" dia). Only the main section of the shot cloud has been considered.

2. Results

2.1 Results Summary

Figure B1 shows the results obtained from Lead No. 3 shot, 36g load (0.030" choke). In Figure B1 the flight time data per range is shown, along with the corresponding values for the velocities, energy per pellet, and shot cloud length. These are quoted for the leading and trailing edges of the main section of the shot cloud only.

Range (m)	Flight Time (ms)		Velocity (m/s)		Energy per Pellet (J)		Shot Cloud Length ¹ (m)
	Leading Edge	Trailing Edge	Leading Edge	Trailing Edge	Leading Edge	Trailing Edge	
20	60.8	68.4	293	250	8.0	5.9	1.9
25	78.7	89.2	265	230	6.6	4.9	2.5
30	98.4	111.8	242	213	5.5	4.2	2.9
35	120.0	136.2	223	198	4.6	3.7	3.3
40	143.3	162.4	206	185	4.0	3.2	3.6
45	168.4	190.3	192	173	3.4	2.8	3.9
50	195.3	220.0	180	163	3.0	2.5	4.1

Figure B1: Shows the flight times for the main section of the cloud, and the corresponding outputs produced by the analysis program for Lead No. 3 shot, 36g load (0.030" choke).

Figures B2, B3, B4, B5 show the variation of flight time, velocity, energy per pellet, and shot cloud length with range, for Lead No. 3 shot, 36g load (0.030" choke). Only the main section of the shot cloud has been considered.

¹ quoted when the leading edge of the main cloud is at the range shown.

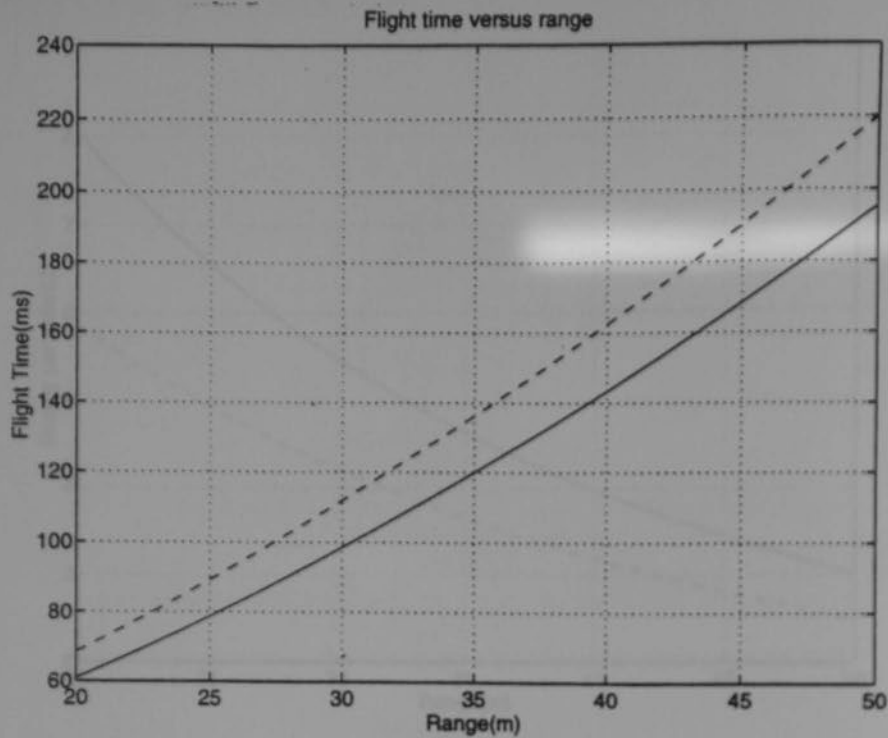


Figure B2: Variation of flight time with range, for the *leading edge* (-) and *trailing edge* (- -) of the main section of the shot cloud, for Lead No. 3 shot, 36g load (0.030" choke).

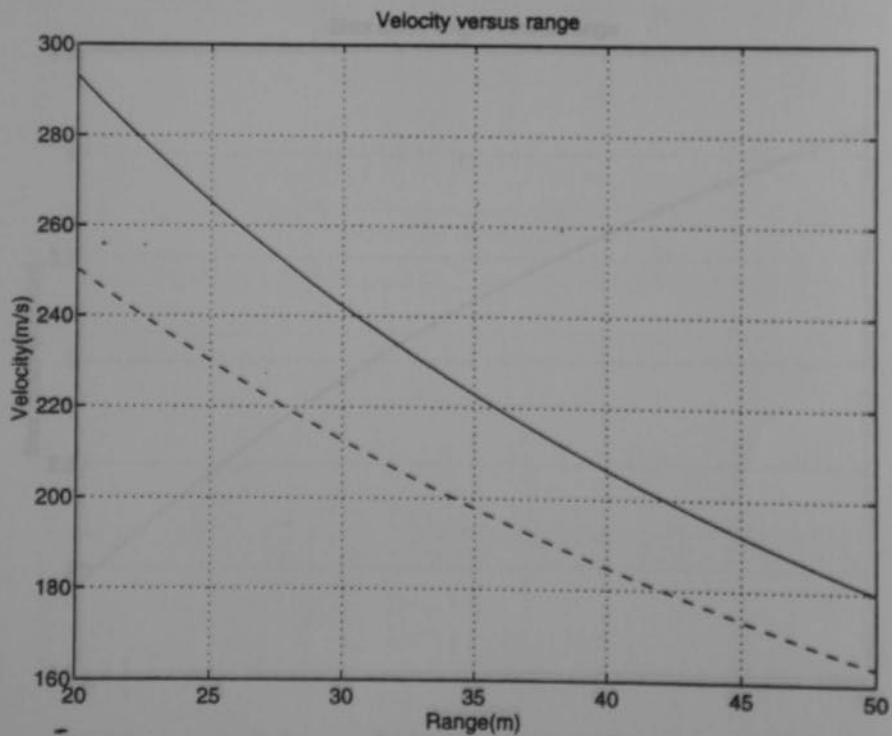


Figure B3: Variation of velocity with range, for the *leading edge* (-) and *trailing edge* (- -) of the main section of the shot cloud, for Lead No. 3 shot, 36g load (0.030" choke).

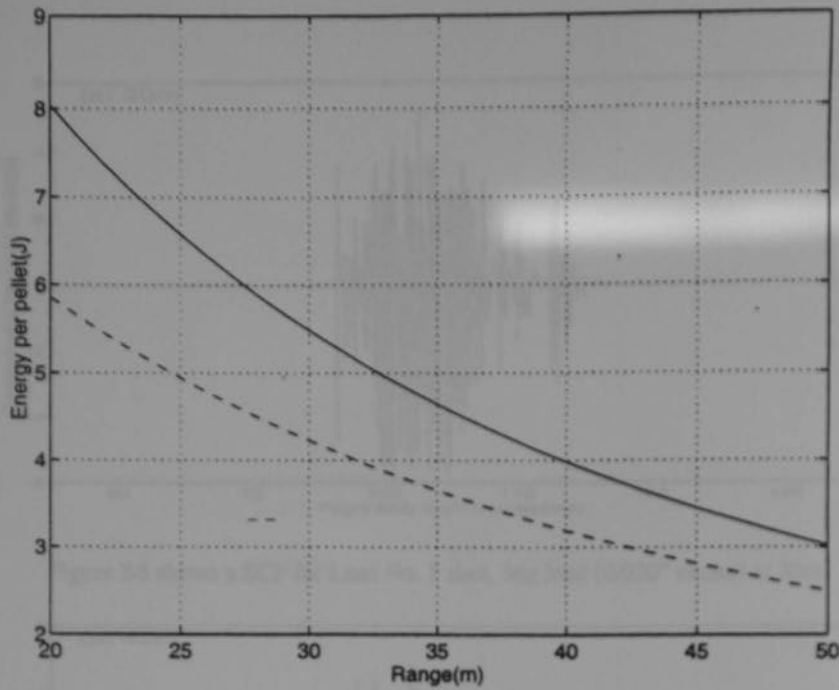


Figure B4: Variation of energy per pellet with range, for the *leading edge* (-) and *trailing edge* (- -) of the main section of the shot cloud, for Lead No. 3 shot, 36g load (0.030" choke).

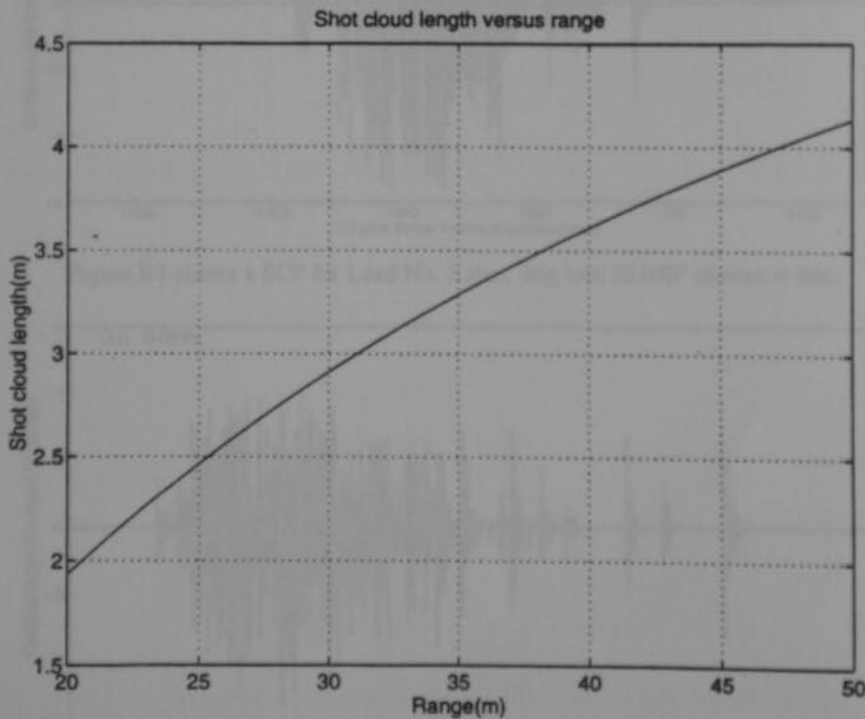


Figure B5: Variation of shot cloud length of the main section of the shot cloud, for Lead No. 3 shot, 36g load (0.030" choke).

2.2. Shot Cloud Profiles (SCP)

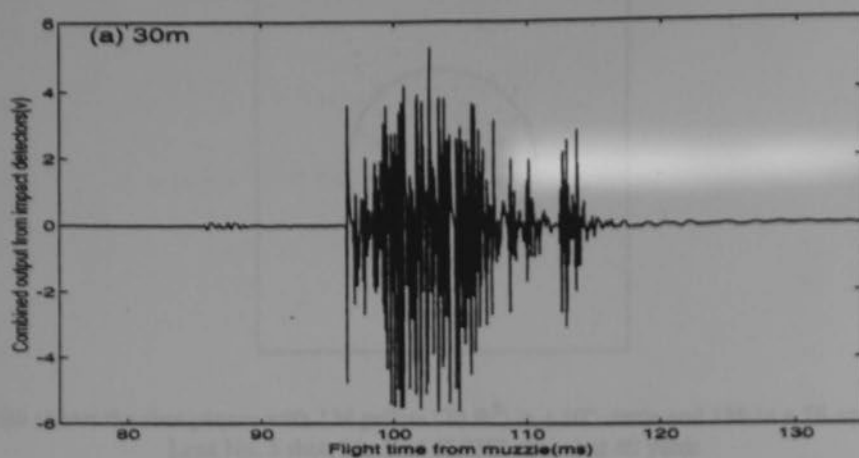


Figure B6 shows a SCP for Lead No. 3 shot, 36g load (0.030" choke) at 30m

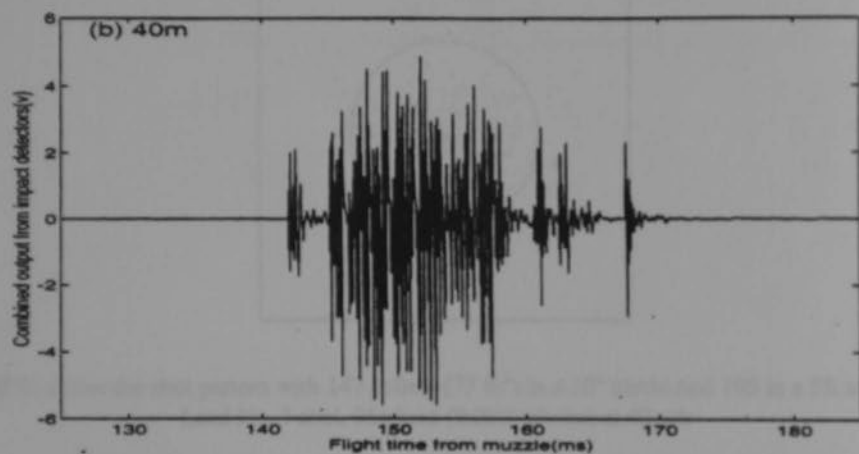


Figure B7 shows a SCP for Lead No. 3 shot, 36g load (0.030" choke) at 40m

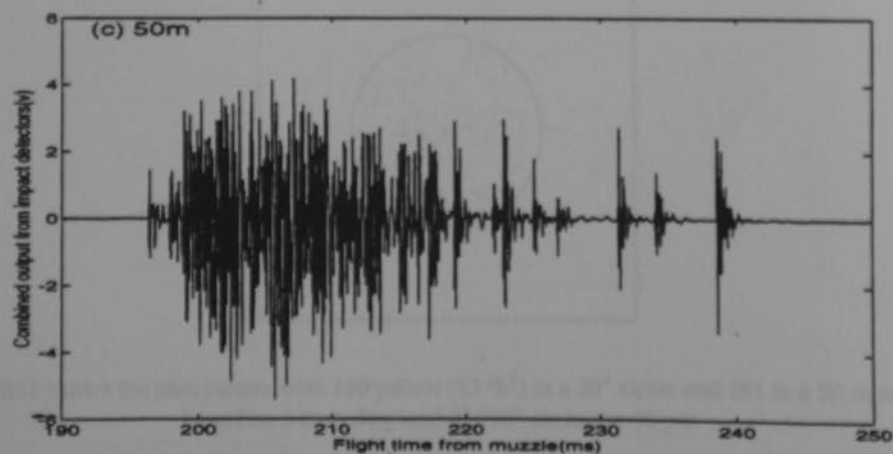


Figure B8 shows SCP for Lead No. 3 shot, 36g load (0.030" choke) at 50m

2.3. Patterns at 40 yards

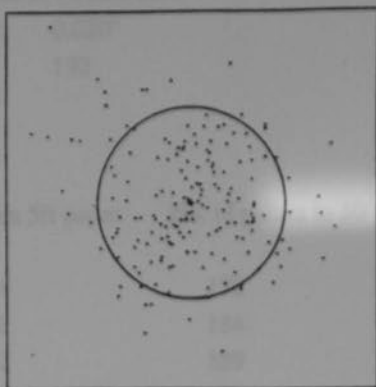


Figure B9 shows the shot pattern with 134 pellets (70 %²) in a 30" circle and 186 in a 5ft square, for Lead No. 3 shot, 36g load (0.030" choke) at 40 yards

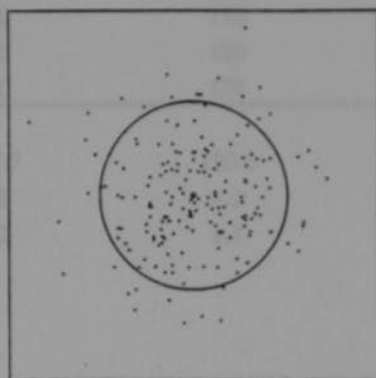


Figure B10 shows the shot pattern with 147 pellets (77 %²) in a 30" circle and 195 in a 5ft square, for Lead No. 3 shot, 36g load (0.030" choke) at 40 yds

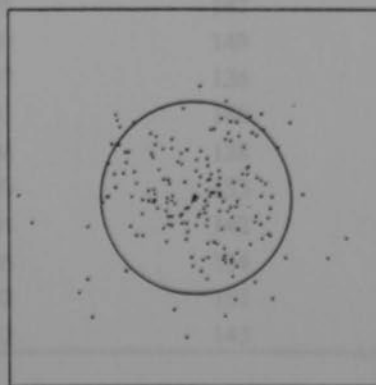


Figure B11 shows the shot pattern with 160 pellets (83 %²) in a 30" circle and 191 in a 5ft square, for Lead No. 3 shot, 36g load (0.030" choke) at 40 yds

² Percentage of the average number of pellets in the cartridge

APPENDIX B

Date: Manufacturer:
 Ammunition: 36g load of #3 lead shot
 Material: Lead Choke: 0.030"
 Average pellets per cartridge: 192

30 yards

Ref.	No. of pellets on 5ft paper	No. of pellets in 30" circle	% in 30" circle
1	184	176	92
2	192	184	96
3	192	189	98
4	187	181	94
5	190	184	96
6	189	181	94
7	192	178	93
8	186	178	93
9	185	180	94
10	180	174	91
Average:	187.7	180.5	94
Std.Dev:	3.8	4.2	2.2
sig/avg: (%)	2.0	2.3	2.3

40 yards

Ref.	No. of pellets on 5ft paper	No. of pellets in 30" circle	% in 30" circle
11	195	147	77
12	189	149	78
13	197	136	71
14	191	160	83
15	196	138	72
16	189	153	80
17	194	142	74
18	186	134	70
19	192	151	79
20	182	143	74
Average:	191.1	145.3	76
Std.Dev:	4.5	7.8	4.1
sig/avg: (%)	2.3	5.4	5.4